



Predicting Antimicrobial Resistance and Empiric Treatment Failure in Hospital-Onset Sepsis Using Machine Learning on Electronic Health Records

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INTRODUCTION

BACKGROUND

- Antimicrobial resistance (AMR) is a leading public health concern, particularly in the context of hospital-acquired infections.
- Delays in appropriate antibiotics increase morbidity and mortality, however studies assessing burden and prognostic impact of antibiotic therapy in this critically ill population are limited.
- Given the limitations of randomized trials, innovative strategies using real-world data and computational methods are needed.

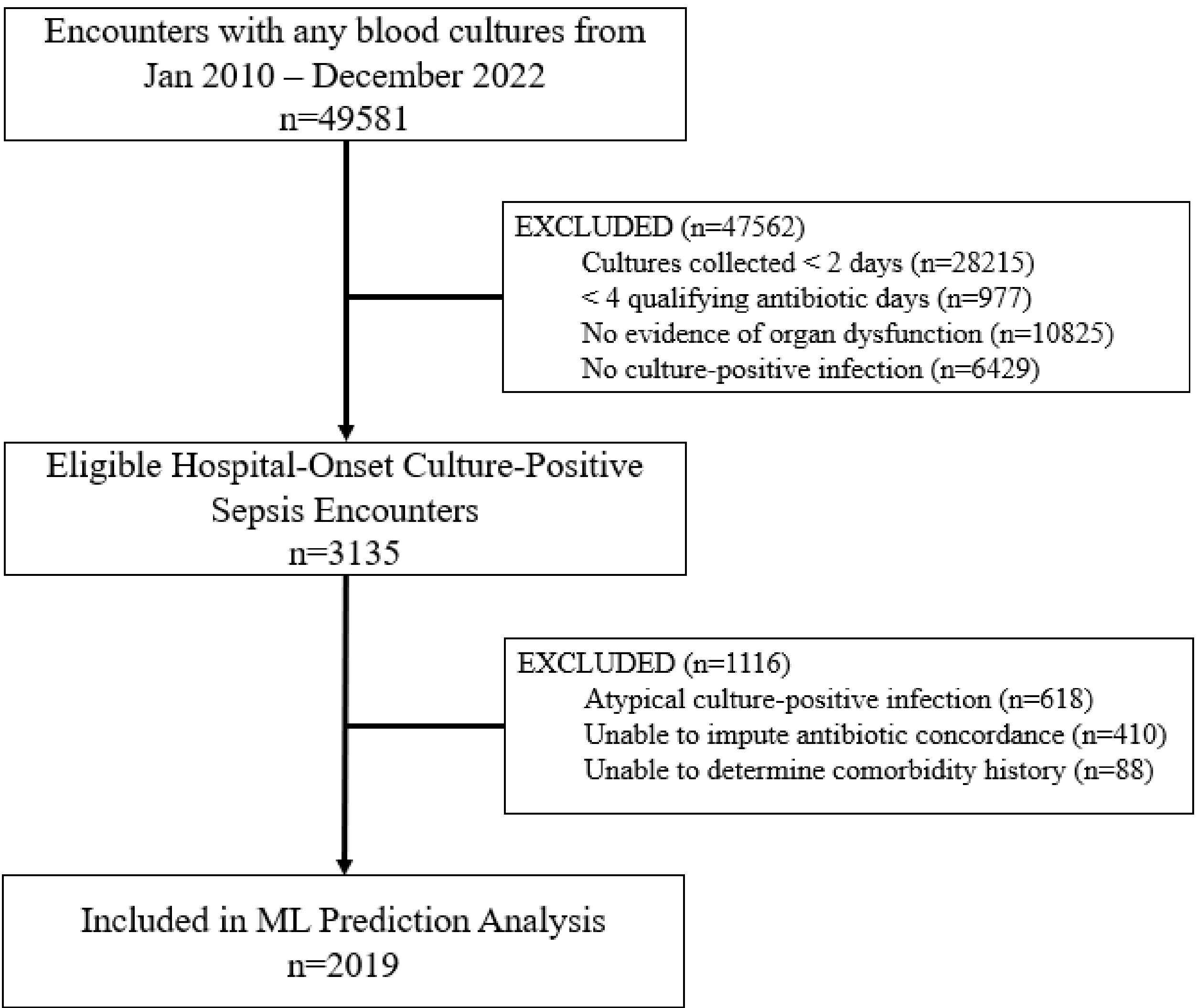
OBJECTIVE:

- We utilized machine learning to predict empiric treatment failure and AMR patterns within a high-risk hospital-onset sepsis cohort using electronic health records (EHRs).

POPULATION OVERVIEW

STUDY DESIGN

- Retrospective cohort study designed using electronic health record data from UF Integrated Data Repository (IRB201900652).
- Approximately 140,000 hospital encounters with documented bacterial infection from 2010 to 2020 were examined from eligibility for adult patients who were hospitalized hospital-onset sepsis.



Resistance Definitions:

- Empiric therapy was considered adequate if all pathogens isolated were susceptible to antimicrobials administered.
- AMR patterns included methicillin (MRSA), extended-spectrum beta-lactamase (ESBL), vancomycin (VRE), ceftriaxone, and carbapenem (CRE).

PREDICTION MODEL DETAILS

Prediction Model Details:

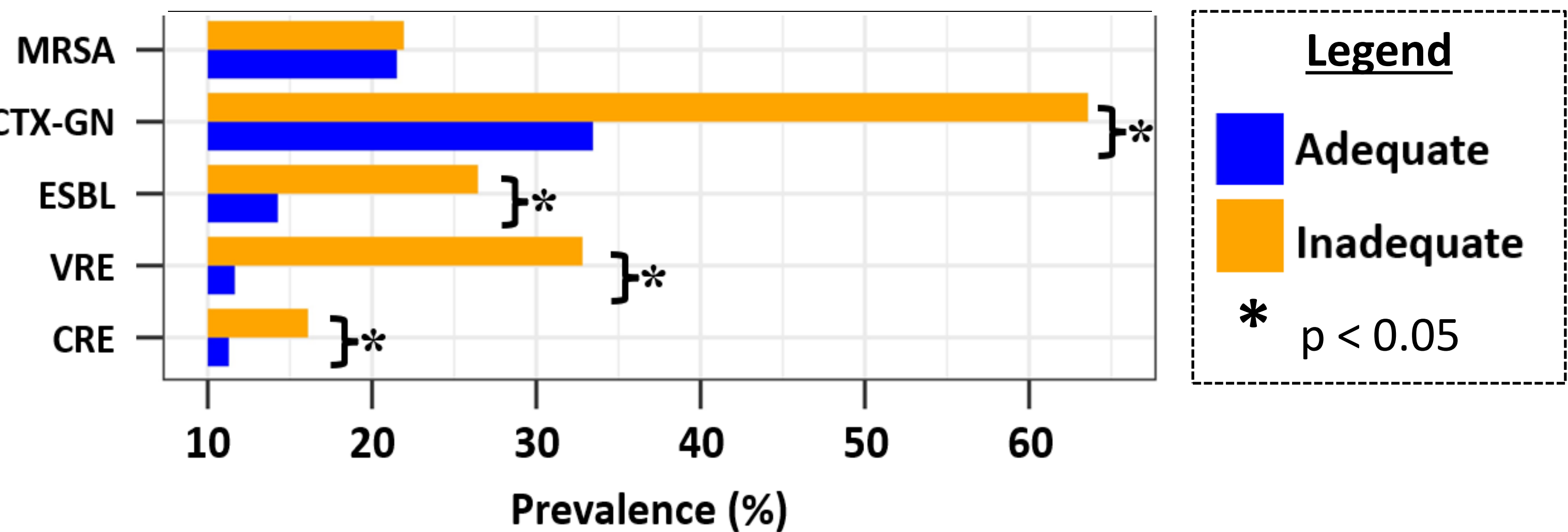
- Features for prediction were patient characteristics, hospitalization details, and clinical severity metrics measured within 24 hours prior to sepsis onset.
- We fit the following linear and nonlinear models: simple logistic regression, boosted logistic regression (BLR), random forest (RF), AdaBoost (AB), support vector machines (SVM), and decision tree (DT).
- We performed bootstrap validation (n = 15) and assessed the out-of-bag for testing. Performance measured using F1 score, precision, recall, specificity, at 0.5 probability threshold, as well as the area under the receiver operating characteristic curve (AUROC).

RESULTS: COHORT COMPARISONS

Table 1: Descriptive Summary by Antimicrobial Resistance Indicator

	No AMR (n=1108)	AMR (n=911)	P-Value
Demographics			
Age (years), mean (SD)	65.6 (16.5)	65.0 (16.1)	0.478
Female – n (%)	509 (45.9%)	391 (42.9%)	0.189
Race, Black – n (%)	260 (23.5%)	248 (27.2%)	0.058
Comorbidity, CCI – n (%)			
Previous myocardial infarction	161 (14.5%)	180 (19.8%)	0.002
Peripheral vascular disease	389 (35.1%)	368 (40.4%)	0.017
Diabetes with complications	174 (15.7%)	206 (22.6%)	<.001
Renal disease	286 (25.8%)	348 (38.2%)	<.001
Congestive heart failure	341 (30.8%)	354 (38.9%)	<.001
Cerebrovascular disease	232 (20.9%)	263 (28.9%)	<.001
Chronic pulmonary disease	412 (37.2%)	412 (45.2%)	<.001
Hemiplegia or paraplegia	59 (5.3%)	88 (9.7%)	<.001
ICU Admission – n (%)	860 (77.6%)	742 (81.4%)	0.039
Infection Source – n (%)			
Intra-Abdominal	225 (20.3%)	185 (20.3%)	1
Bone/Joint	59 (5.3%)	70 (7.7%)	0.039
Pulmonary	727 (65.6%)	655 (71.9%)	0.003
Skin/Soft Tissue	131 (11.8%)	141 (15.5%)	0.02
Inadequate Empiric – n (%)	66 (6%)	311 (34.1%)	<.001
30-day mortality – n (%)	252 (22.7%)	279 (30.6%)	<.001

Figure 2: Prevalence of resistance class by empiric antibiotics at 24 hours



RESULTS: PREDICTION MODELS

Table 2: Area Under ROC curve for antimicrobial resistance pattern and inadequate empiric treatment by machine learning model

	DT	BLR	RF	AB	SVM
Any AMR					
	0.58	0.63	0.64	0.62	0.62
	(0.58-0.59)	(0.62-0.64)	(0.63-0.64)	(0.61-0.63)	(0.61-0.63)
CRE					
	0.55	0.57	0.70	0.65	0.64
	(0.50-0.59)	(0.54-0.60)	(0.67-0.73)	(0.61-0.68)	(0.62-0.66)
ESBL					
	0.53	0.60	0.59	0.58	0.58
	(0.52-0.55)	(0.57-0.63)	(0.57-0.61)	(0.57-0.59)	(0.56-0.60)
MRSA					
	0.57	0.65	0.66	0.63	0.66
	(0.54-0.59)	(0.62-0.69)	(0.65-0.67)	(0.61-0.64)	(0.65-0.67)
VRE					
	0.59	0.67	0.71	0.71	0.69
	(0.57-0.62)	(0.62-0.72)	(0.69-0.72)	(0.69-0.73)	(0.68-0.71)
Inadequate Empiric					
	0.58	0.66	0.65	0.65	0.66
	(0.56-0.60)	(0.64-0.67)	(0.64-0.66)	(0.64-0.66)	(0.65-0.67)

Table 3: Classification metrics for best for best performing model. Metrics are optimized to maximize sensitivity and specificity.

	Sensitivity	Specificity	F1	Precision
Any AMR				
	0.65	0.57	0.64	0.50
	(0.61-0.69)	(0.53-0.62)	(0.63-0.64)	(0.48-0.51)
CRE				
	0.80	0.59	0.13	0.10
	(0.73-0.87)	(0.51-0.67)	(0.11-0.15)	(0.07-0.14)
ESBL				
	0.61	0.57	0.19	0.15
	(0.51-0.72)	(0.46-0.67)	(0.17-0.20)	(0.12-0.18)
MRSA				
	0.60	0.67	0.30	0.23
	(0.55-0.66)	(0.61-0.72)	(0.28-0.32)	(0.21-0.25)
VRE				
	0.71	0.64	0.23	0.21
	(0.64-0.77)	(0.57-0.71)	(0.21-0.25)	(0.17-0.25)
Inadequate Empiric				
	0.64	0.62	0.39	0.29
	(0.58-0.69)	(0.56-0.68)	(0.38-0.40)	(0.27-0.31)

CONCLUSION

- In a hospital-onset sepsis cohort with prevalent AMR, machine learning approaches utilizing limited EHR data accessible at initial sepsis recognition had moderate, but clinically insufficient, discriminatory ability in predicting AMR and empiric antimicrobial treatment failure.
- Future research will evaluate the utilization of large language models using unstructured EHRs.